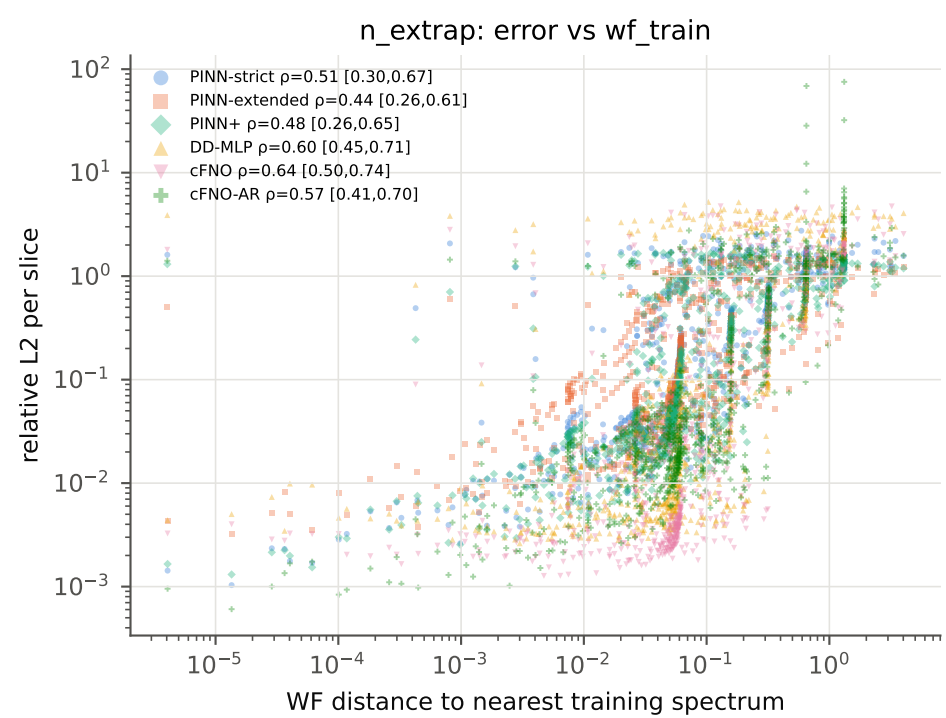
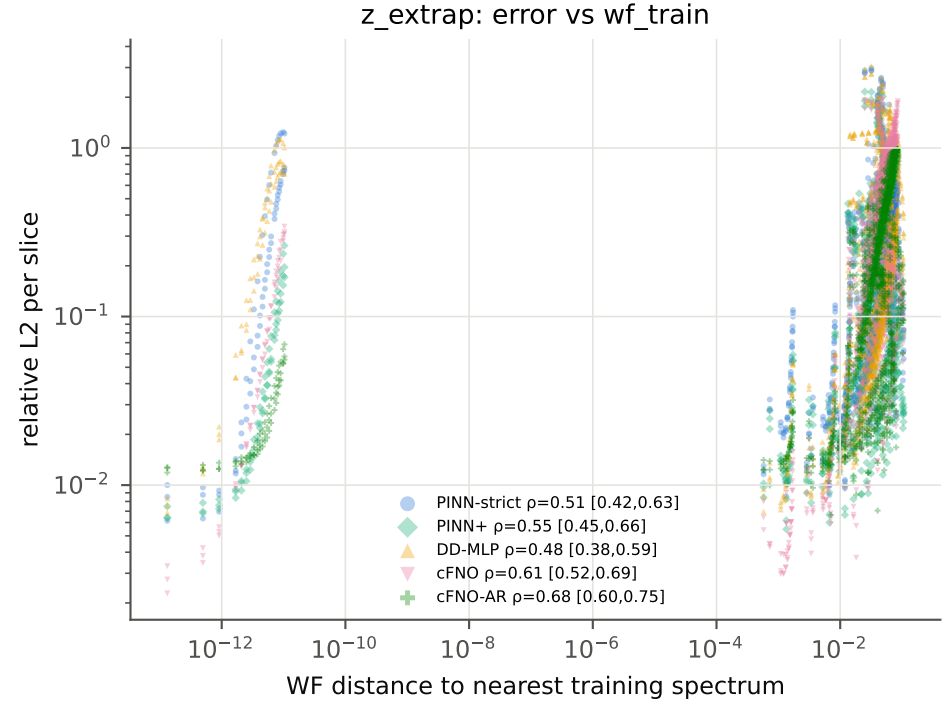

n_extrap: error vs wf_train
PINN-strict ρ=0.51 [0.30,0.67]
PINN-extended ρ=0.44 [0.26,0.61]
PINN+ ρ=0.48 [0.26,0.65]
DD-MLP ρ=0.60 [0.45,0.71]
cFNO ρ=0.64 [0.50,0.74]
cFNO-AR ρ=0.57 [0.41,0.70]
relative L2 per slice
WF distance to nearest training spectrum


n_extrap: error vs phi_excess
PINN-strict ρ=0.66 [0.49,0.77]
PINN-extended ρ=0.57 [0.40,0.70]
PINN+ ρ=0.62 [0.46,0.74]
DD-MLP ρ=0.70 [0.53,0.80]
cFNO ρ=0.71 [0.54,0.82]
cFNO-AR ρ=0.69 [0.52,0.80]
relative L2 per slice
Δφ = 0: inside the training box
SPM phase beyond training box, Δϕ


z_extrap: error vs wf_train
PINN-strict ρ=0.51 [0.42,0.63]
PINN+ ρ=0.55 [0.45,0.66]
DD-MLP ρ=0.48 [0.38,0.59]
cFNO ρ=0.61 [0.52,0.69]
cFNO-AR ρ=0.68 [0.60,0.75]
relative L2 per slice
WF distance to nearest training spectrum


z_extrap: error vs phi_excess
PINN-strict ρ=0.75 [0.71,0.81]
PINN+ ρ=0.90 [0.88,0.93]
DD-MLP ρ=0.59 [0.54,0.65]
cFNO ρ=0.92 [0.89,0.94]
cFNO-AR ρ=0.87 [0.83,0.91]
relative L2 per slice
Δφ = 0: inside the training box
SPM phase beyond training box, Δϕ